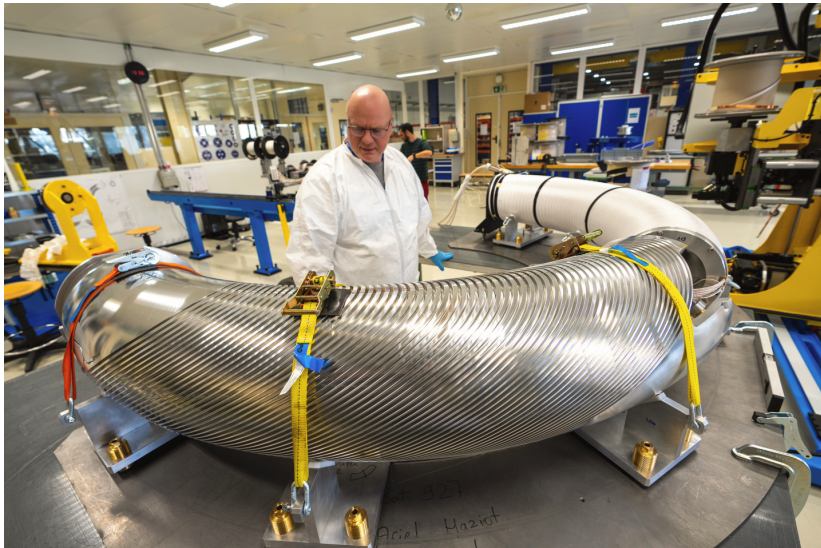


A curved magnet with much potential

Rope winding and giant pasta shapes meet particle physics in Fusillo, in a new magnet prototype with possible medical applications in hadron therapy



Frédéric Garnier controls the process of inserting the inner former into the outer former. The grooves in the outer former, into which the cable was later fed, are clearly visible. (Image: M. Struik/CERN)

What do rope winding and giant pasta shapes have to do with particle physics? The answer is a new superconducting magnet prototype under development at CERN, lovingly named Fusillo because of its shape. Originally, CERN physicists became interested in developing this technology for use in compact particle accelerators, for example in a new storage ring for the CERN-based experiment [ISOLDE](#). However, development of this technology could also have a big impact in the medical field. For example, one of the possible future applications of magnets like Fusillo is in hadron therapy to treat cancer.

Hadron therapy is a type of radiotherapy that uses beams of protons or light ions to irradiate cancer tissue. Compared to X-rays, which use beams of light, beams of ions release less energy along their path and more energy in one specific spot. This causes less radiation damage to the healthy tissue surrounding the tumour and allows a higher dosage to be administered safely, resulting in a faster destruction of the tumour. Less damage to the surrounding tissue also leads to a lower toxicity rate, meaning the patient will feel better during and after treatment.

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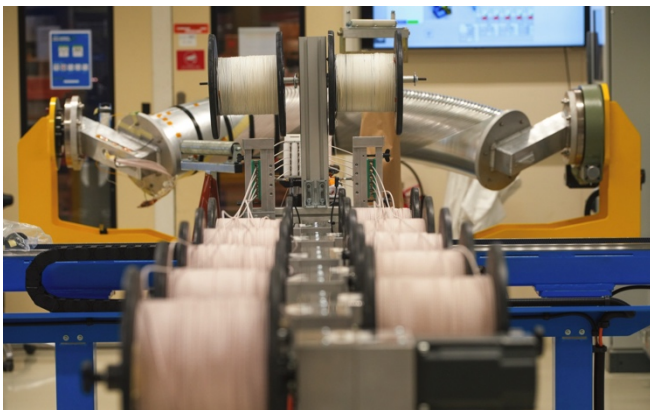
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Paul Kuijter (1957 – 2025)
Senamile Masango (1987 – 2025)

In the last few decades, over a hundred hadron therapy facilities have been built worldwide. However, few medical centres have the means to acquire the machines for this treatment as the magnets needed are costly and typically require resources like high currents and helium cooling. So far, hadron therapy is only offered in some European and Asian countries and the USA, while there are none in Africa and only one under construction in South America. With the technology of Fusillo, a Curved-Canted-Cosine-Theta (CCCT) dipole demonstrator, the treatment might become more accessible in the future. This new type of electromagnet has many benefits: it requires a much lower current, has a relatively low cost and is more compact than other magnets due to a simplified design that requires fewer individual parts. Fusillo is also expected to be more easily cooled by dry cooling without using liquid helium, unlike many other superconducting magnets.



Ten ropes, each made of seven insulated wires, are combined into a cable before being wound around the metal former that can be seen in the background. (Image: M. Struik/CERN)

The basis of Fusillo's design is cable wound into two nested coils. The coils are tilted following the grooves of the so-called "former", seen in the first picture. The inner coil is tilted in the opposite direction to the outer coil, and together they produce a dipole field inside the tube. This is a new approach to creating a dipole field, which is being tested by the Fusillo demonstrator. Although the underlying idea has been around for many

decades, the necessary computing power for its design has only been available in recent years. To produce a magnet that can be powered by a low current, Ariel Haziot and his team at CERN decided to twist multiple insulated wires into a type of rope that is then wound around the former. The individual wires are connected in a way that allows the current to run many laps around the coils, creating an electromagnet of usable strength (3 T in the centre) that requires a relatively small amount of current (300 A).

The Canted-Cosine-Theta (CCT) concept has been in development at CERN since 2014 and is used in the High-Luminosity LHC. The current project of building a curved CCT demonstrator has so far taken Haziot and his team about 2.5 years. The photos document the recent process of winding the cables around the former. With only a couple of final steps to go, the building phase is drawing to a close and the first full-scale tests of this new magnet are planned for April. Following several subscale assessments, the tests will compare the magnet against simulation results and determine the next steps in the development process. Magnets like Fusillo are expected to be in use in the new storage ring for HIE-ISOLDE within about five years and might then be developed further for other applications, including hadron therapy.



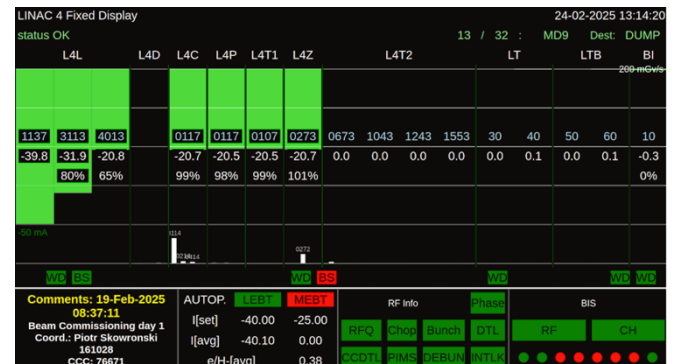
Carlos Fernandes (left), Frédéric Garnier (centre) and Ariel Haziot (right) wind the ropes of cable around the former. (Image: M. Struik/CERN)

Insa Meinke

Accelerator Report: Linac4 beam commissioning is nearing completion and the PS Booster is next in line

While the first H^- ion beams have been accelerated up to 160 MeV in Linac4, the PS Booster is preparing for its first proton injection of 2025

On 19 February, as planned, the beam stoppers just downstream of the Linac4 ion source were opened, allowing the first H^- beam of 2025 to travel into the Linac4 accelerator. This marked the official start of its beam commissioning, which is a meticulous process of carefully guiding the beam through the linear accelerator while fine-tuning all its parameters. To do this, the Linac4 team adjusts the voltages and phases of the high-frequency power waves that travel through the accelerating structures. These waves gradually increase the energy of the H^- ions, ensuring that, by the time they reach the end of Linac4, they have been successfully accelerated to an energy of 160 MeV. As I write, the hardware re-commissioning of the PS Booster (PSB), which began on 13 February, is progressing well and remains on schedule – the injection of the first 2025 proton beam into the PSB is planned for 27 February. From that point onward, Linac4 should be fully operational, providing beams to the PSB in a stable and reliable way, ready for an efficient PSB beam commissioning.



The Linac4 display showing the beam as it passes through the accelerating structures. The green bar on the left-hand side shows the H^- ion beam current (-40 mA) out of the source into Linac4. After some initial losses, the beam is accelerated to 160 MeV and sent to the beam dump at the end of Linac4, located just before the transfer line that connects Linac4 with the PSB. This is why the green bars end at L4Z. (Image: CERN)

Meanwhile, on 22 February, the hardware re-commissioning of both the Proton Synchrotron (PS) and the Super Proton Synchrotron (SPS) also began. The two accelerators are on track to receive their first proton beams and begin beam commissioning on 5 March and 14 March, respectively.

Rende Steerenberg

From H^- ions in Linac4 to protons in the PS Booster

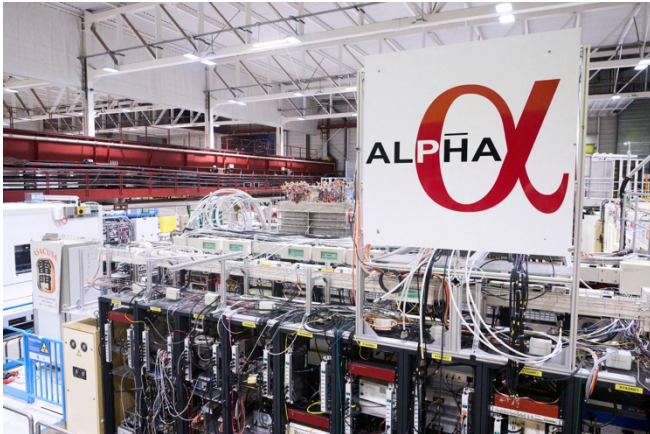
When charged particles pass through a magnetic field, their direction of motion depends on their charge. Negatively charged particles, such as the H^- ions produced in the Linac4 source, bend in the opposite direction to positively charged particles, such as the protons used in our downstream accelerators. Linac4 accelerates H^- ions to 160 MeV before steering them toward the PS Booster (PSB) injection region using magnets. At this point, the H^- ions pass through a thin carbon foil, which strips away their two electrons. This process, known as charge exchange injection, leaves behind only protons, which are then guided into the PSB ring using additional magnets.

As these protons complete their first orbit inside the PSB, Linac4 continues to deliver fresh H^- ions. Since H^- ions and protons bend in opposite directions in the same magnetic field, they naturally mix just before reaching the carbon foil. This means that both the protons already circulating in the PSB and the incoming H^- ions pass through the foil, where the H^- ions lose their electrons and turn into protons, becoming part of the already circulating beam.

This process is repeated multiple times until enough protons have been accumulated in the PSB. At this point, the PSB switches off the magnetic fields guiding protons through the foil, ensuring that no extra particles pass through unnecessarily. The accumulated protons are then accelerated and sent further downstream in the accelerator chain. This method of injection allows for efficient multiturn accumulation, leading to higher beam intensities and higher beam densities, which will ultimately provide a large number of collisions in the LHC.

Optical fibre link to make CERN more on time than ever

A new optical fibre link between CERN and Paris will provide the Laboratory with an accurate frequency reference, enhancing precision and supporting experiments like ALPHA in their search for matter–antimatter differences

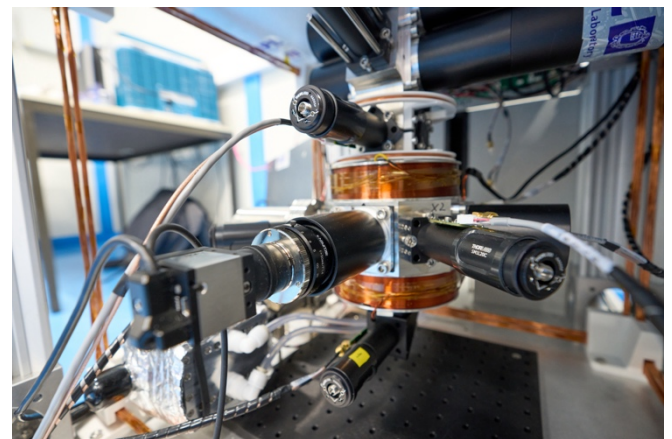


An optical fibre link to the ALPHA experiment at CERN's Antimatter Factory will assist in precise measurements of the properties of antihydrogen. (Image: CERN)

The Standard Model of particle physics is based on the idea that if you simultaneously swap a matter particle for its antimatter version (changing the sign of its charge), flip its spatial coordinates as if viewed through a mirror (parity) and reverse the direction of time, then there should be no difference in the behaviour of the two particles. Due to the central role of fundamental symmetry in quantum field theory, discovering even a small violation of this principle, known as charge-parity-time (CPT) symmetry, would suggest that our understanding is incomplete and could point to new physics beyond the Standard Model. Experiments at CERN's Antimatter Factory test fundamental principles such as CPT symmetry by studying the properties and behaviour of antimatter and comparing them with normal matter. The [ALPHA](#) experiment performs such tests through spectroscopy of antihydrogen – that is, by measuring the frequencies of transitions in the anti-atom using laser light or microwaves. If the results match those of normal hydrogen, the measurement is consistent with CPT symmetry. These frequencies, measured in units of Hz, equivalent to one per second, correspond to the energy level intervals in atoms and the spectral lines that arise when they make quantum transitions between levels. To accurately compare matter and antimatter, the frequencies must be determined incredibly precisely, requiring ultra-

precise clocks. That's why a caesium fountain clock was recently installed in ALPHA and a new optical fibre link between the experiment and the French National Metrological Institute in Paris is now online. Both the clock and the optical link will help improve the precision of ALPHA's antihydrogen measurements by orders of magnitude.

"For our [previous measurement](#) of the transition between the ground state and the first excited state of antihydrogen, we used a simpler clock made out of a quartz oscillator referenced via GPS satellite as a frequency reference, and we reached a precision on the transition frequency of two parts per trillion (2×10^{-12})," says physicist Janko Nauta from the ALPHA collaboration. "However, the equivalent measurement on hydrogen, performed a [few years before our antihydrogen measurement](#), has an even higher precision, of four parts per quadrillion (4×10^{-15}), calling for a better clock to look for potential differences between matter and antimatter."



The SI second is defined as the duration of 9 192 631 770 oscillations between two levels of the ground state of the caesium-133 atom. The caesium fountain clock that the ALPHA researchers received and installed in 2022 tells them exactly how long one second lasts. (Image: CERN)

"For ALPHA, both the optical fibre link and the caesium fountain clock play important roles in making antihydrogen measurements with a precision that matches that of the hydrogen measurements," continues Nauta. "While we rely on the clock, the link helps us to reduce noise in

the measurement and to better evaluate the clock in the long term, to verify that it stays accurate. In addition, the link will make it possible to use signals from optical quantum clocks in the future, surpassing the stability of clocks that currently realise the SI second.”

The link is part of the [REFIMEVE](#) network, a project that distributes an ultra-stable optical frequency reference to research laboratories across France and beyond via existing optical cables on the French internet network. It is a pilot implementation of a new project that aims to connect multiple experiments at CERN to REFIMEVE. This has the potential to improve the precision of clocks across CERN and could provide a new way for the Laboratory to access Coordinated Universal Time (UTC) – the global standard for timekeeping. The optical signal from the link can synchronise with UTC more precisely

than via GPS satellite, which is currently used across CERN.

“The [CERN Quantum Technology Initiative](#) envisages having more precise frequency signals delivered to CERN from other national metrology institutes and distributing them to all the interested experiments at the Laboratory,” says Edoardo Martelli from CERN’s IT department. “Having more sources allows a more precise synchronisation of the local clocks and increases the robustness of the service.”

ALPHA’s [most recent precise measurement](#) of the transition between the ground state and the first excited state of antihydrogen has placed tighter constraints on violations of CPT symmetry than its previous measurement. With the new optical link, the collaboration hopes to put CPT symmetry to even more stringent tests.

Naomi Dinmore

The Prime Minister of Luxembourg visits CERN



His Excellency Mr Luc Frieden, Prime Minister of the Grand Duchy of Luxembourg, signs CERN’s guestbook in the presence of the Director-General, Fabiola Gianotti, on 26 February 2025. (Image: CERN)

On 26 February, CERN welcomed His Excellency Mr Luc Frieden, Prime Minister of the Grand Duchy of Luxembourg. The Prime Minister and his

delegation were received at Point 1 of the LHC by CERN’s Director-General, Fabiola Gianotti, the Director for Research and Computing, Joachim Mnich, the Director for International Relations, Charlotte Warakaulle, the Head of the Information Technology department, Enrica Porcari, the Head of Relations with Associate and non-Member States, Emmanuel Tsesselis, and the Senior Advisor on Relations with Luxembourg, Christoph Schäfer.

Accompanied by the spokesperson-elect of the ATLAS collaboration, Stéphane Willocq, the delegation then visited the ATLAS control room and experiment cavern.

CERN’s formal collaboration with Luxembourg began in October 2024, with [the signature of an agreement for a new partnership](#) funded by the Government of Luxembourg, the [United Nations World Food Programme](#) (WFP), the [Luxembourg Institute of Science and Technology](#) (LIST) and CERN. The aim of this collaboration is to use expertise in AI technology to address the issue of global hunger.

Anaïs Schaeffer

International Geneva celebrates the International Year of Quantum

Hosted in the historical venue of the Geneva Conservatory of Music, the event brought quantum technologies on stage for the local and global community



Panel including representatives of permanent missions in Geneva (Italy, Mexico, Kenya, Vanuatu, Slovenia, India and the Kingdom of Saudi Arabia) exploring the key themes of quantum science and technology and its impact on humanity. (Image: CERN)

On 21 February 2025, the Geneva Conservatory of Music welcomed over 100 members of the International Geneva community to celebrate the launch of the [International Year of Quantum Science and Technology \(IYQ\)](#), following its official inauguration at [UNESCO](#) Headquarters in Paris earlier this month.

Co-organised by the [Open Quantum Institute](#) (OQI) at CERN and UNESCO, the launch of the IYQ in Geneva was an officially recognised global IYQ event, designed to raise awareness within the local international community of policymakers, diplomats, scientists and civil society of the societal impact of quantum technology.

International Geneva is a renowned global hub for international scientific diplomacy and therefore serves as the ideal platform for shaping a shared vision for quantum technology development and ensuring that its benefits reach communities all over the world, particularly in under-resourced

regions. Key themes explored during the celebration included equitable access to quantum technology, open science, capacity building, education and bridging the digital divide.

“International cooperation is at the heart of what we do at CERN,” emphasised Enrica Porcari, Head of CERN’s IT department, during her opening remarks. “In particular, with our OQI programme, we collaborate with a range of stakeholders from academia, the private sector, diplomacy, industry and education to anticipate the impacts of the technology before they are realised and to ensure that it is developed for the benefit of all.”

Part of the collaborative dialogue between OQI and UNESCO, the Geneva launch of the IYQ underscored the need for global collaboration and anticipating the multilateral governance structure that will be needed to develop sustainable and inclusive quantum technology.

Throughout the year, OQI at CERN will lead the organisation of key events and activities worldwide, including regional events, hackathons and [the Quantum Diplomacy Game](#), which will be played across several continents. At CERN, a dedicated events season will allow local audiences to dive deep into the quantum world and appreciate how fundamental research can foster societal impact through its technological implementations.

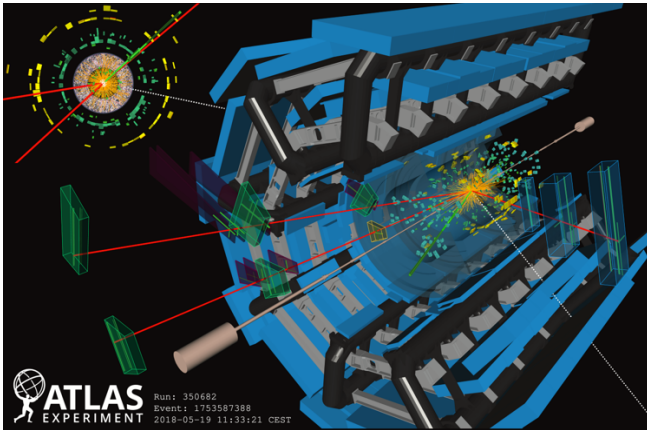
More information about CERN events can be found at: <https://visit.cern/events>

As an officially recognised global event of the IYQ, this event was made possible thanks to the support of all [IYQ partners](#).

Alexia Yiannouli

Observing triplets of weak bosons

ATLAS reports the first observation of a Z boson accompanied by two other carriers of the weak force



A three-vector-boson event recorded by ATLAS, with one W boson decaying to an electron and a neutrino, one W boson decaying to a muon and a neutrino, and a Z boson decaying to two muons. Muons are shown by red lines, the electron by a green line, and “missing energy” from the neutrino by a dashed white line. (Image: ATLAS/CERN)

As the carriers of the weak force, the [W](#) and [Z](#) bosons are central to the [Standard Model](#) of particle physics. Though discovered four decades ago, the W and Z bosons continue to provide physicists with new avenues for exploration. In a [new study](#), the [ATLAS](#) collaboration analysed its full data set from the second run of the [Large Hadron Collider](#) (recorded from 2015 to 2018) in search of a rare process in which a Z boson is produced alongside two other weak-force carriers, or vector (V) bosons, as the W and Z are known.

“The production of three vector bosons is a very rare process at the LHC,” says Fabio Cerutti, ATLAS Physics Coordinator. “Its measurement provides information on the interactions among multiple bosons, which are linked to underlying symmetries of the Standard Model. This is a powerful tool for uncovering new physics phenomena, such as new, undiscovered particles that are too heavy to be directly produced at the LHC.”

The ATLAS team observed “VVZ” production with a statistical significance of 6.4 standard deviations, surpassing the [five standard deviation](#) threshold needed to claim observation. This process has only 1 in 100 000 000 000 chances of occurring and is hard to isolate from those that mimic its signatures. The observation extends previous results from the ATLAS and CMS collaborations, including the [observation of VVV production by CMS](#) and the [observation of WWW production by ATLAS](#).

As some of the heaviest known particles, the W and Z bosons can decay in a myriad of different ways. In the [new ATLAS study](#), physicists focused on seven “decay channels” with the highest discovery potential. These channels were further refined using a machine-learning technique called boosted decision trees, with algorithms trained for each channel to identify the sought-after signal. The successful observation showcases the meticulous work of the analysis team, especially given the complexity of the decay combinations. By combining the decay channels, ATLAS researchers were able to observe VVZ production and set limits on contributions from new-physics phenomena to the signal. The resulting limits confirm the validity of the Standard Model and are in line with previous results on the production of three vector bosons.

As researchers analyse larger data sets from the LHC’s third run and the upcoming [HL-LHC](#), they will further refine their measurements of the production of three vector bosons – deepening our understanding of these fundamental particles and their role in the Universe.

Read more on the [ATLAS website](#).

ATLAS collaboration

Women and Girls in Science and Technology 2025: reaching out to over 4700 pupils



Maria Alandes Pradillo, from CERN's IT department, gives a presentation at the École des Vergers, in Meyrin, in a previous edition of Women and Girls in Science and Technology. (Image: CERN)

From 10 to 14 February, 76 ambassadors from CERN, the University of Geneva (UNIGE), the Swiss Federal Institute of Technology Lausanne (EPFL) and the Annecy Particle Physics Laboratory (LAPP) visited schools in the region to inspire young people and show them that science is accessible to everyone. Some 215 presentations were given to approximately 4730 pupils aged from 5 to 16.

The aim of this programme, which is organised as part of the [International Day of Women and Girls in Science](#), held on 11 February this year, is to combat gender stereotypes in scientific and technical careers. Since its inception in 2017, the initiative has given more than 30 000 pupils the opportunity to meet female researchers, engineers, technicians and other professionals working in STEM fields (science, technology, engineering and maths) and learn more about their careers and their work.

The volunteers shared their passion for science through interactive presentations adapted to each age group, experiments, demonstrations and enriching discussions. The teachers emphasised the positive impact of these interactions, which make it easier for pupils, especially girls, to picture themselves pursuing scientific careers.

What pupils had to say:

"If my boyfriend enters a black hole, can I still phone him?"

"Is the speed of light faster than a cheetah?"

And their teachers:

"Our speaker explained to the children that she was dyslexic and that there might be mistakes in her slides. This shows pupils who have learning difficulties that they needn't be an obstacle to a career in science!"

"In one class, the pupil who was struggling the most has found her vocation. Her attitude to lessons has changed now that she's understood the value of studying. You've helped her find her calling."

Would you like to volunteer to share your passion for science with the next generation? Contact us to find out about our upcoming calls for volunteers: education.locale@cern.ch.

Lou Planchamp

A bestiary of exotic hadrons

The LHC is engaged in a complex quest to understand the exotic offspring of one of the fundamental forces of nature

The [Large Hadron Collider](#) (LHC) has discovered 76 new particles so far. In addition to discovering the unique [Higgs boson](#), the world's largest and most

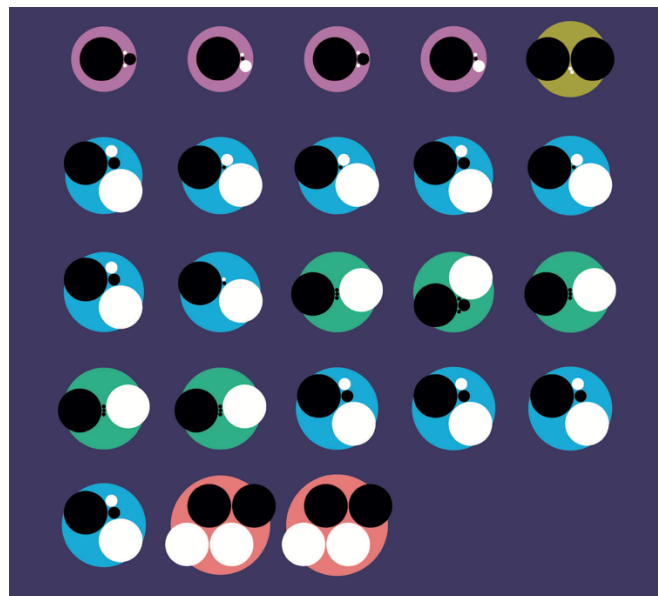
powerful particle accelerator has found 52 new hadrons made up of two or three elementary particles called quarks and a bestiary of 23 oddities

which confound easy explanation. These exotic hadrons appear to be composed of four or five quarks, but how they stick together is a subject of hot debate.

The November/December 2024 edition of [CERN Courier](#) attacks the question on two fronts.

As experimental particle physicists from CERN's [LHCb](#) experiment, Patrick Koppenburg of Nikhef and Marco Pappagallo of INFN and the University of Bari approach the question as a [detective story](#) – an experimentally driven quest to understand the exotic offspring of one of the fundamental forces of nature.

Meanwhile, theorists Marek Karliner of Tel Aviv University and Jonathan L. Rosner of the University of Chicago offer piercing insights into the [quantum anatomy of exotic hadrons](#), revealing them to behave at times like molecules, at times as novel compact states, and at times as both simultaneously – with much still to be discovered.



A simplified visualisation of the 23 exotic hadrons discovered so far at the LHC. Each particle contains at least one charm quark and a mixture of up, down and strange quarks. Quark and hadron masses are indicated by area, and quarks and antimatter quarks are shown in black and white, respectively. See [CERN Courier](#) for a full explanation. (Image: M. Rayner)

Find out more in the two *CERN Courier* articles [here](#) and [here](#).

CERN Science Gateway welcomes its 500 000th visitor



CERN Science Gateway passed the 500 000 visitor mark on Friday, 14 February with the visit of Damian (left) and Saira (right). (Image: CERN)

[CERN Science Gateway](#) has welcomed its 500 000th visitor, just 16 months after it opened to the public. Since October 2023, the Laboratory's

education and outreach centre has been a must-visit for tourists to Geneva, showcasing the science, discoveries and technology of CERN in an interactive way, housed in a unique building designed by renowned architect Renzo Piano.

Damian (left) and Saira (right) took a two-day trip from Zurich to Geneva, with an obligatory stop at Science Gateway. He is an architecture student, she studies mathematics, and both were excited to discover CERN.

As a thank you treat for their visit, Damian and Saira enjoyed a meal at the Big Bang Café and received the book edited by the Renzo Piano Foundation about the construction of CERN's science and outreach centre.

Want to discover all that Science Gateway has to offer the CERN community? [Become a CERN guide](#) or [visit](#) the education and outreach centre from Tuesday to Sunday.

CERN invites photographers to explore the colliders of the future

On 27 May 2025, twenty photographers will be given an exclusive opportunity to explore the world's largest particle physics laboratory



A photographer in front of concrete blocks in the antimatter hall, pictured by Cédric Favero. Photowalk 2018, 1st prize. (Image: Cédric Favero / CERN)

Cameras at the ready! On 27 May 2025, twenty photographers will be given an exclusive opportunity to explore the world's largest particle physics laboratory. CERN is teaming up with major science laboratories for an international photography competition known as the Global Physics Photowalk.

During this year's competition at CERN, entitled the "CERN Photowalk 2025: Future Colliders", the Laboratory will open the doors to four unique

spaces. The photographers will discover the facilities of the High-Luminosity LHC, its new magnets and cold powering system, and some of the technologies that will be needed for the machines of the future in decades to come.

The best photos will be chosen by a local jury and a public vote. Each laboratory will also choose three photos to be entered into the international competition, which will run at the end of the summer. The winning photos in the international competition will be selected by a jury and a public vote and will be featured in the [CERN Courier](#) and in [Symmetry magazine](#).

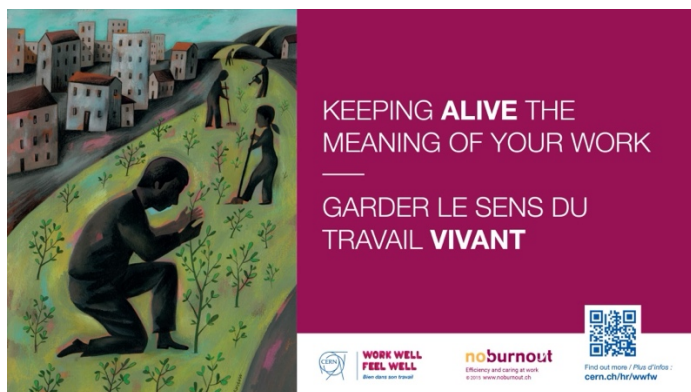
The "CERN Photowalk 2025: Future Colliders" competition is open to all photographers. For more information about it, including how to enter, please go to: <https://photowalk.web.cern.ch>. Please note that the number of participants is limited and that the deadline for applications is 23 March 2025, 23:59:00 CET.

For more information about the international competition, "Global Physics Photowalk 2025", please go to [this link](#).

Follow #PhysPics25 on social media for the latest news about this event.

Work Well Feel Well: Keeping alive the meaning of your work

Part nine of the [Work Well Feel Well](#) series looks at how to our core values help to keep us motivated



Aligning with our core values

Our core values form the basis of our motivation each day, often without us realising. When we feel that our actions are useful and consistent with who we are, it gives them meaning and helps us to rise to challenges. Psychologists refer to this as "intrinsic motivation".

In the course of our professional lives, we change and so do our work environments. So, it can be useful to periodically reflect on our situation,

looking beyond our immediate day-to-day responsibilities.

- “What does “success” mean for me?”
- “What are the core values that motivate me?”

We need to balance the meaning that we give to our profession with the day-to-day realities of our job. If we don’t link our ideals to day-to-day constraints, our ideals can work against us, creating an internal conflict.

- “Once conscious of my core values, how can I express them?”
- “How can I enrich my work by reframing tasks to align with my core values?”

Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now

offers useful resources that can be downloaded, including:

- an [exercise](#) to reflect on what you like about your job
- an [exercise](#) to explore the meaning behind day-to-day tasks
- CERN’s [Code of Conduct](#), which outlines the Organization’s core values – integrity, commitment, professionalism, creativity and diversity – upon which our competency model is built. Our personal core values may align with these and extend beyond them. Many lists of values exist online and can be useful starting points to reflect on our own values.

HR department

Computer Security: 2025 – Plugging holes

[CERN’s computing infrastructure is vast and heterogeneous, dynamic and complex. Keeping it secure is a marathon](#) – trying to stay ahead of new deployments in order to guarantee decent protection, but often just running behind, trying to regain control. The Computer Security Office has been attempting to plug obvious holes in CERN’s computer security stance since well before the [2023 cybersecurity audit](#), and it will continue to do in 2025 and beyond.

Very early measures to maintain control were already established in the 2000s, when CERN’s outer perimeter firewall was toggled from “mostly open” to “closed with controlled openings”. Since then, all incoming accesses have required [a dedicated opening](#), while outgoing traffic is by default blocked on all so-called “lower” ports (i.e. ports 0-1023/tcp and udp). In doing so, CERN regained visibility and control over what kind of traffic enters and leaves CERN, providing proper protection and early detection. One hole plugged. Also in the 2000s the Technical Network (TN) was isolated from what was then called the General Purpose Network (GPN), such that only explicitly permitted hosts could communicate between both worlds. While that provided a first level of

control and protection, its “trusted/exposed”-mechanism based on simple IP addresses and Access Control Lists (ACLs) remained coarse, vague and still too open. This is why Long Shutdown 3 (LS3) will see a replacement of “trusted/exposed” by a fully fledged redundant pair of firewalls that are better able to filter and select between desirable and undesirable traffic. In the course of their deployment, all central IT services and some control systems will need to neatly define which parts (e.g. “front-end”) to open between the networks – a restriction based on need, not on convenience. Another hole plugged.

Similarly, today’s Campus network (i.e. large parts of the aforementioned GPN, hosting office PCs and the wireless network) will be separated from CERN’s data centre networks via a similar set of firewalls. As for the TN, only necessary servers and services will be exposed to CERN IT’s clients, while the “internal goings-on” will stay within the data centres and no longer be accessible to third parties. Hole three plugged.

Turning to CERN’s web sphere, its “internet presence”, many different penetration tests have [revealed time and again weaknesses, misconfigurations and vulnerabilities](#). Still, today,

the choice of which website is publicly visible and which is not lies with the webmaster, uncontrolled and unreviewed by the Computer Security Office. With the deployment of a Web Application Firewall, a layer of monitoring and protection will be inserted in order to catch at least the basic blunders and mistakes. Mandatory “[Security Principles](#)” will help website owners to further improve the security stance of their sites. Two more holes plugged.

While [a new mail application](#) already plugged additional holes linked to malicious emails, phishing attempts and the like, the 2024 deployment of antispooofing measures (“SPF/DKIM/DMARK”) and the upcoming deployment of “impersonation protections” will further protect CERN mailboxes from malicious evildoers. With that, we are done. Several holes, plugged in one go.

Finally, accounts. In February 2025, a two-year-long campaign to [deploy 2-factor authentication](#) (2FA) to all CERN primary and secondary accounts concluded. At least it concluded for the CERN Single Sign-On, now protecting access to many CERN websites. A big hole plugged. And further holes will be plugged with the roll-out of 2FA to CERN’s interactive Linux service (“LXPLUS”) and Windows Terminal Service

(“CERNTS”). In addition, discussions have started on how to improve the security of so-called service accounts (2FA, too?) and how to better distinguish between secondary accounts (alternative accounts for humans) and service accounts (intended to be used by automatisms). While this is still pending, CERN’s password rules will evolve this year, dropping the requirement for “a mix of capital and small letters, numbers and symbols” and instead simply [requiring a passphrase of 15 characters](#): “Fais de ta vie un rêve, et d’un rêve, une réalité”, “In Xanadu did Kubla Khan a stately pleasure dome decree!”, and even “Mmm Mmm Mmm Mmm Mmm” will become acceptable passwords. The hole created by password rules that are too weak should be plugged soon.

Of course, it is natural that there are more holes and new holes will follow. With the [Computer Security Board](#) now in place, the existing and new rules that are subsidiary to the [CERN Computing Rules](#) will be officially approved, as future rules will be too, helping to control CERN’s vast and heterogeneous, dynamic and complex computing landscape, providing additional layers of protection, and helping to plug the (security) holes of the future.

Computer Security Office

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Safety update: key findings and actions following the 17 January incident

Following the [Safety incident on 17 January](#) that occurred in Building 182 dedicated to Neutrino Platform (NP) activities and involving two members of the CERN personnel, a *Major Safety Incident Analysis Group* (MSIAG) was created as per [Specific Safety Instruction, SSI-SIM-2-1](#) to establish all relevant facts and possible causes.

Their conclusions and preliminary recommendations were submitted to the CERN Director-General on 7 February. A Major Safety Incident Board (MSIB) was then convened, which reviewed the Major Safety Incident Analysis Report produced by the MSIAG. The MSIB’s final ‘Incident Board report’, detailing conclusions and

recommendations based on this review, was endorsed by the Director-General on 24 February. As a result of their analysis, the MSIB made 24 actionable recommendations that further strengthen established safety practices in place at CERN. The majority of these focus on immediate actions required before any further NP-related activities can resume. These include reaffirming the ownership of the NP laboratory's infrastructure and equipment and the associated roles and responsibilities in safety management and reviewing reviewing risk assessments and resulting working procedures to enhance their

adaptability to activities that constantly evolve over time. Several recommendations further extend CERN-wide, specifically reinforcing workplace induction at locations with specific hazards and refining associated safety protocols. The MSIB report will now be communicated to the recipients as defined in the SSI-SIM-2-1, which includes the recommendation owners.

In line with this SSI, CERN will ensure the prompt and effective follow up of the recommended actions for continued safe operations and collaboration in the NP laboratory and, more generally, the Organization as a whole.

Taxation in Switzerland

Memorandum concerning the internal tax annual certificate, the individual annual statement and income tax declaration forms in Switzerland for the reference year 2024

I – Internal tax annual certificate and individual annual statement for 2024

Your internal tax annual certificate or individual annual statement for 2024, issued by the Finance and Administrative processes Department, is available via [MyFiles](#) (under « Financial and Social Benefits »). The document that you have received (certificate or statement) depends on your situation at CERN in 2024. It is intended exclusively for the tax authorities.

1. If you are currently a member of the CERN personnel, you have received an e-mail containing a link to your certificate or statement, which you can print out if necessary.
2. If you are no longer a member of the CERN personnel or are unable to access your

certificate or statement as indicated above, you will find information explaining how to obtain one [here](#).

II – 2024 income tax declaration forms in Switzerland

The Admin e-guide can be found [on this page](#) to provide further general indications for completing the 2024 income tax declaration form, and to offer support in this matter.

If you have any specific questions, please contact your tax office directly.

NB: The information regarding the French income tax declaration form is usually available in April.

Contact: HR-Internal-tax@cern.ch

HR department

Summer work for children of members of the personnel

During the period from 16 June to 5 September 2025 inclusive, there will be a limited number of jobs for summer work at CERN (normally unskilled work of routine nature), which will be made available to children of members of the personnel (i.e. anyone holding an employment or association

contract with the Organization). Candidates must be aged between 18 and 24 inclusive on the first day of the contract and must have insurance covering illness.

Candidates must already hold a residence permit granted by one of CERN's host states (France or

Switzerland) at the time of their application and must have a valid residence permit. Proof will be required.

The duration of all contracts will be 4 consecutive weeks and the allowance will be CHF 1587 for this period. Candidates should apply via HR department's electronic recruitment system, SmartRecruiters: <https://smrtr.io/pvMNV>

Completed application forms must be returned by 16 March 2025 at the latest. To allow as many people as possible to benefit, each child may participate in this programme only once.

The results of the selection will be available mid of May 2025.

If you have any questions, please contact us [here](#).

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

3 Mar 15:00 | Experiments | Main Auditorium and online | LHC Experiment Committee (LHCC) | EN
[161st LHC Experiment Committee \(LHCC\) meeting - open session](#)

4–6 Mar | Computing | Council Chamber | CERN openlab | EN
[2025 CERN openlab Technical Workshop](#)

7 Mar 09:30 | Knowledge Sharing | 30/7-018 - Kjell Johnsen Auditorium | KT Seminars | EN
[A Brief History of Time: From High-Energy Physics to AI-Driven Cancer Care](#)

7 Mar 11:00 | Knowledge Sharing | Online and Salle Bohr (40/S2-B01) | CERN Alumni | EN
[CERN Alumni Virtual Company Showroom with Transmutex](#)

11 Mar 17:30 | Knowledge Sharing | Main Auditorium | Staff Association | FR
[Excellence at work, as seen through the eyes of neuroscientists](#)

14 Mar 14:30 | Knowledge Sharing | Online | CERN Alumni | EN
[Moving Out of Academia to Data Science & Analytics \(2nd edition\)](#)

20 Mar 20:00 | Knowledge Sharing | CERN Science Gateway | Public Events | FR
[Les Dieux et le Père Noël : vers une solution quantique](#) – part of CERN's [Quantum!](#) season

21 Mar 14:00 | Knowledge Sharing | Online | Micro-Talks | EN
[#15 Critical Thinking: Challenge Assumptions, Reframe Issues](#)

26 Mar 16:00 | Knowledge Sharing | 40/S2-C01 - Salle Curie | Early career community | EN
[LHC Soft Skills Workshop: Presenting Science -- How to prepare a scientific presentation with impact!](#)

Registration now open

20–23 May | Engineering | CERN | FPGA Developers' Forum | EN
[2nd FPGA Developers' Forum \(FDF\) meeting](#)

28 May | Knowledge Sharing | EPFL, Lausanne | Famelab | EN, FR or DE
[Famelab Switzerland 2025 local heats, open to 18–35-year-olds](#)

15–27 Jun | Accelerators | Bulgaria | CERN Accelerator School | EN
[Pushing the Limits: Intensity Limitations in Hadron Beams](#)

23–27 Jun | Physics | Venice, Italy | Update of the European Strategy for Particle Physics | EN
[Open Symposium of the 2026 Update of the European Strategy for Particle Physics](#)

6–19 Jul | Computing | Lund, Sweden | CERN School of Computing | EN
[46th CERN School of Computing \(CSC 2025\)](#)

This is a curated list of events relevant to the CERN community. More events are available [here](#):

home.cern/events

Indico also shows ALL events [happening today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch.

Private parties in Restaurant 1 – Terms and conditions

Note that only **authorised** private parties can take place in Restaurant no.1. The authorisation has to be sought, at least 48 hours in advance, through [CERN Service Portal: Authorisation request to organise a drink in the Restaurant 1 extension](#).

The terms and conditions are the following:

- [Operational Circular No.08 Dealing with alcohol-related problems](#) shall be observed.
 - Private events shall not disturb activities of the Organization in any respect. Any noisy disturbance has to be avoided after 22:00.
 - The party organiser bears the responsibility for the event and for security reasons, shall be on site for the duration of the event.
 - All participants have to observe [CERN Code of Conduct](#) in every respect.
 - The party shall be stopped at the latest at 23:30 (except for Saturdays and Sundays 21:30)
 - The furniture (tables, chairs, etc.) shall not be moved; any damages shall be borne by the organiser
 - Immediately after the event, the cutlery, plates and glasses shall be cleared and put back at the right place. The tables and the place in general shall be cleaned properly. The waste shall be put in the appropriate bins. Glass bottles shall be removed, no glass recycling being available at the restaurants. Any additional cleaning to be provided by the restaurant staff will be charged to the organiser.
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Vaccination and duty travel

A vital measure to prevent disease and protect individual and collective health

Vaccination remains one of the most effective tools in public health. According to the [World Health Organization](#) (WHO), every year vaccination helps prevent between 3.5 and 5 million deaths from avoidable diseases including measles, pneumonia and rotavirus-induced diarrhoea. It has eradicated smallpox, drastically reduced polio cases and helps protect vulnerable populations by maintaining high levels of vaccination coverage. However, declining vaccination rates have led to the resurgence of diseases such as whooping cough and influenza,

underscoring the importance of mass immunisation efforts.

In professional settings, vaccination serves a dual purpose: protecting employees from work-related risks and preventing disease spread within organisations. This contributes to collective immunity, safeguarding the entire workforce.

Duty travel and vaccination

If you are travelling for work, it is important to be aware of the specific health risks linked to your destination. If you are an employed (MPE) or associated (MPA) member of the personnel, we

highly recommend you make an appointment well in advance – ideally six weeks before your departure – with CERN’s occupational health doctor for a duty travel consultation. This will provide you with an opportunity for a general health check and to get vaccination recommendations tailored to your destination. Members of the personnel also benefit from a travel assistance provided by International SOS when going on duty travel. This assistance, both medical and security related, can be used anytime before, during or after the trip through its [website](#) or its [mobile app](#).

Note that some specific vaccinations, like for yellow fever, have to be administered in registered centres, such as the [Tropical and Humanitarian Medicine Division](#) at the *Hôpitaux Universitaires de Genève* (HUG). You should also consult such a

centre if you have a particular vaccination status or needs.

You can contact the CERN Medical Service to book an appointment for work-related travel vaccinations by phone: +41 22 76 73186 or email: medical.service@cern.ch.

For more information, please refer to the following webpages:

- [Vaccine Scheduler | ECDC](#)
- <https://www.unicef.org/immunization>
- [Vaccinations en santé au travail – Article de revue – INRS](#) (in French)
- [EpiScope – janvier 2025 – Nouveautés 2025 des déclarations de maladies transmissibles | ge.ch](#) (in French)

HSE unit

Obituaries

Paul Kuijer (1957 – 2025)



(Image credit: Nikhef)

We received news of the passing of Paul Kuijer with great sorrow. Paul was a senior experimental physicist at Nikhef and played an important role in the ALICE experiment at CERN. During the last years of his career, Paul had turned his attention mostly towards the ET Pathfinder project within the Gravitational Waves group at Nikhef and the University of Maastricht. While he officially retired

in 2024, he still came to Nikhef weekly and continued to contribute to several projects.

Paul worked on his PhD at the University of Amsterdam with the MARK-J experiment at the PETRA accelerator in DESY, Hamburg, where he performed a search for the top quark and measured the strong coupling constant using electron-positron collisions. In 1987, he joined Utrecht University as an assistant professor and worked on various experiments at the Institute for Nuclear Physics. In 1994, he became a member of the ALICE experiment at CERN and worked on the proposal and technical design report for the Silicon Strip Detectors (SSD), a joint project realised with laboratories in Finland, France, Italy, The Netherlands, Russia and Ukraine. In 2008, he became the first ALICE run-coordinator and, in 2009, he was selected as deputy spokesperson of the ALICE collaboration.

When his role as deputy spokesperson came to an end, he again worked on silicon detectors at Nikhef, overseeing the maintenance of the SSD,

and, until 2019, was the project leader for the upgrade of the new ALICE Inner Tracking System. In addition to working on hardware, Paul had a strong passion for physics, supervising many PhD students throughout his career. Everyone who worked with him remembers him fondly and with respect. He was an excellent scientist and a gentle, reliable person who would always make time to discuss physics and help with the daily practical problems encountered by technicians, PhD students or staff members. Paul was known for his hands-on approach; he often had a small project

on the go, and he never shied away from new challenges. He was very approachable and always made time to discuss physics or simply chat with students and staff alike.

We will greatly miss his friendly and warm personality. Paul leaves a great void in our community. Our thoughts are with the family, friends and close colleagues that Paul leaves behind.

His friends in the ALICE collaboration

Senamile Masango (1987 – 2025)



During her master's studies at the University of the Western Cape in South Africa, Senamile Masango came to CERN in 2017 as part of an African-led team conducting an experiment at the ISOLDE experiment facility. (Image: CERN)

It is with a heavy heart that we share the news of the passing of Senamile Masango, a member of the CERN Alumni Network. A pioneering South African nuclear scientist, entrepreneur and advocate for the UN Sustainable Development Goals, she dedicated her life to advancing science and empowering women in STEM. As the founder of the Senamile Masango Foundation, she worked to build a brighter future for Africa, ensuring that no one was left behind in the pursuit of knowledge and opportunity.

Hailing from a Zulu royal lineage, Senamile began university at the age of 16. During her master's studies at the University of the Western Cape (UWC) in South Africa, she [came to CERN in 2017](#) as the only female member of an African-led team that conducted an experiment at ISOLDE to investigate the isotope selenium 70. She graduated with a master's degree cum laude in nuclear physics from UWC, based on work done at TRIUMF, the Canadian National Facility for Nuclear and Particle Physics.

She was named one of the 50 Global Inspirational Women of 2020, was a Women in Tech Global Awards finalist in 2021 and was a recipient of the International Women in Science Award in 2022.

Senamile was a strong supporter of the [CERN Alumni Network](#), recognising the importance of global scientific collaboration and mentorship. She was a [keynote speaker](#) at CERN Alumni Third Collisions in 2024, with an inspiring talk entitled "Changing the World Through Science by Taking Science to the Community," which deeply resonated with the audience, reflecting her passion for making science accessible to all.

Our thoughts go out to Senamile's family, friends and colleagues.

The CERN Alumni Relations Team